OCTAVIA KGWETE - DATA ANALYTICS WITH SQL

SQL FOUNDATIONS - EXERCISE 02

SQL AGGREGATE FUNCTIONS & OPERATIONS

01
```
SELECT DISTINCT department
FROM students_table;
```
↳

department
IT
Finance
HR

02
```
SELECT AVG(age) AS avg_age,
       department
FROM students_table
GROUP BY department;
```
↳

department	avg_age
IT	21.5
Finance	22

03
```
SELECT COUNT(*) AS student_count,
       AVG(age) AS avg_age,
       department
FROM students_table
GROUP BY department
HAVING COUNT(*) > 1;
```
↳

department	student_count
IT	2
HR	2

04

```
SELECT *
FROM students_table
WHERE age BETWEEN 21 AND 23;
```

↳

student-id	name	age
2	Bob	22
3	Charlie	21
4	Diana	23
5	Eve	22

05

```
SELECT *
FROM students_table
WHERE department IN ('IT', 'HR')
AND age > 21;
```

↳

student_id	name	age	department
4	Diana	23	IT
5	Eve	22	HR

06 SELECT SUM(credits) AS total_credits
department
FROM courses_table
GROUP BY department
HAVING SUM(credits) > 5;

↳

department	total_credits
IT	11

07 SELECT course_id,
course_name,
department,
~~FROM~~ credits
FROM courses_table
WHERE credits <> 4;

↳

course_id	course_name	department	credits
104	Excel	Finance	2
105	Statistics	HR	3
~~105~~101	SQL Basics	IT	3

08 SELECT course_id,
course_name,
credits
FROM courses_table
ORDER BY credits DESC
LIMIT 3;

↳

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3.

06 ~~SELECT SUM(credits) AS total_credits~~

~~FROM courses_table~~

~~GROUP BY department~~

~~HAVING SUM(credits) > 5;~~

```
09 SELECT MAX(grade) AS max_grade,
          MIN(grade) AS min_grade,
          AVG(grade) AS ~~avg~~avg_grade
   FROM   enrollments_table;
```

max_grade	min_grade	avg_grade
90	78	84,6

```
10 SELECT   COUNT(*), AS enrollment_count
            course_id
   FROM     enrollments_table
   GROUP BY course_id
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

```
11 SELECT   department
            SUM(salary) AS total_salary
            SUM(bonus) AS total_bonus
   FROM     salaries
   GROUP BY department;
```

department	total_salary	total_bonus
IT	122 000	10500
HR	109 000	7500
Finance	70 000	6000

12
```
SELECT AVG(salary) AS avg_salary,
       department
FROM   salaries_table
GROUP BY department
HAVING AVG(salary) > 55000;
```

department	avg_salary
IT	61 000
Finance	70 000

13
```
SELECT employee_id,
       name,
       salary,
       bonus
       (salary + bonus) AS total_compensation
FROM   salaries_table
WHERE  salary AND BONUS > 60 000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5000	65000
3	Spike	70 000	6000	76000
4	Tyke	62000	5500	67500

14
```
SELECT  SUM(budget) AS total_budget,
        AVG(budget) AS avg-budget,
        department
FROM    projects_table
GROUP BY department
HAVING  AVG(budget) > 70 000;
```

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15
```
SELECT  project_id
        project_name
        department
        budget
FROM    projects_table
WHERE   budget BETWEEN 50000 AND 120000
  AND   department <> 'Marketing';
```

project_id	project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	HR Portal	HR	50 000